HANDS-ON WRITING A DMP

Faculty of Sciences

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Attribute to: Ghent University Data Stewards

CHECK-IN

What's your research about? Funding?

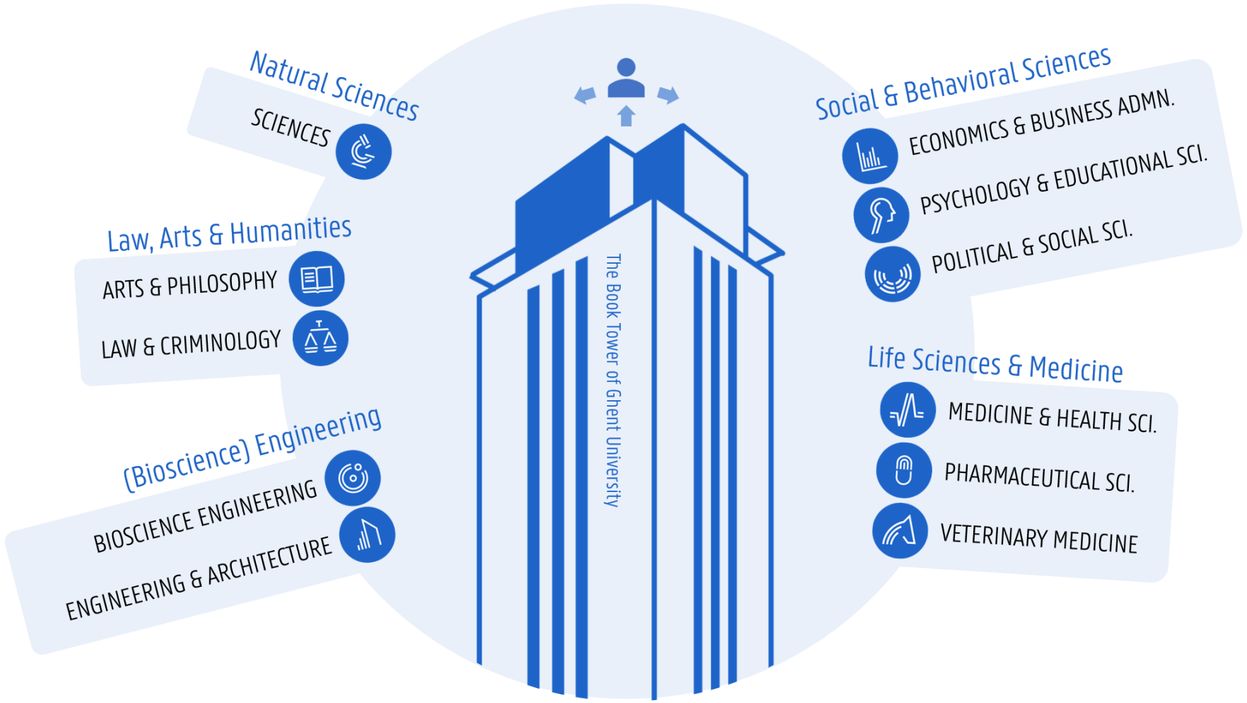
What's the status of your project & your DMP?

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DATA STEWARDS: WHO ARE WE?

1 Between central & faculty level

Team within **Ghent University Library** (Book tower), with one team coordinator and one data steward for each **faculty cluster**:



2 Three main task areas

Advisors, not operational data managers!

1. Raise awareness, inform, train & advise
2. RDM infrastructure-related projects
3. Monitor & shape RDM policy and strategy

HOW CAN WE HELP?

Data Management Plan support
templates, guidance, examples, feedback

Online information & guidance
via RDM webpages, (re)search tips...

RDM helpdesk
for all RDM-related questions

Consultations

Training, info sessions & other events

User support for Library RDM tools
e.g. DMPonline.be, OSF for UGent

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WHAT'S ON TODAY

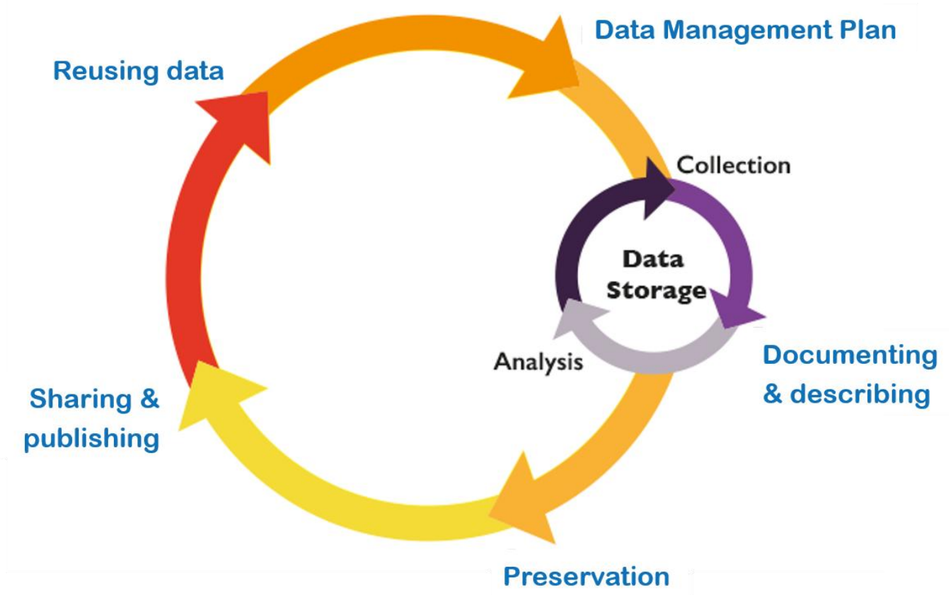
- Update and complete your DMP based on the project's reality using data workflows and data summary tables
- Spot and address data management "red flags" before closing your project
- Prepare your data for sharing and preservation

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DATA MANAGEMENT PLAN (DMP)

- Outlines how research data will be handled **during** & **after** a project.
- Increasingly required by research funders/institutions

  -> as a **first step** towards good **research data management** (RDM)



Adapted from DCC



PUBLICATIONS AND DATA

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DMP SECTIONS



Data summary & description

Purpose, origin, type, sensitiveness, format & volume



Documentation & metadata

Information to keep data findable, understandable, reusable



Roles & responsibilities



Ethical & legal issues

Intellectual property, GDPR, etc.



Long-term preservation

What to preserve & where, for how long, access restrictions



Costs & resources



Data storage & security

Security measures, back-up, data transfers & access control



Data sharing and reuse

Data repositories, access conditions, licenses or agreements

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DATA SUMMARY / DESCRIPTION SECTION

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DMP SECTIONS



Data summary & description

Purpose, origin, type, sensitiveness, format & volume

Documentation & metadata

Information to keep data findable, understandable, reusable



Roles & responsibilities

Describe the data that will be used and/or generated during the research project

Ethical & legal issues

Intellectual property, GDPR, etc.



Long-term preservation

What type, how stored, for how long, access restrictions



Costs & resources

Different data types require different management approaches

First step to identify needs and design data management strategy

Data storage & security

Security measures, back-up, data transfers & access control

Data sharing and reuse

Data repositories, access conditions, licenses or agreements

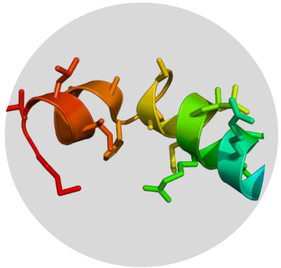
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RESEARCH DATA CAN MEAN A LOT

Survey



Measurement



Observation



Code



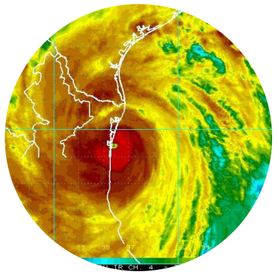
Image



Document



Visualisation



Research data is **any information** that has been collected, observed, generated or created **to validate original research findings**.

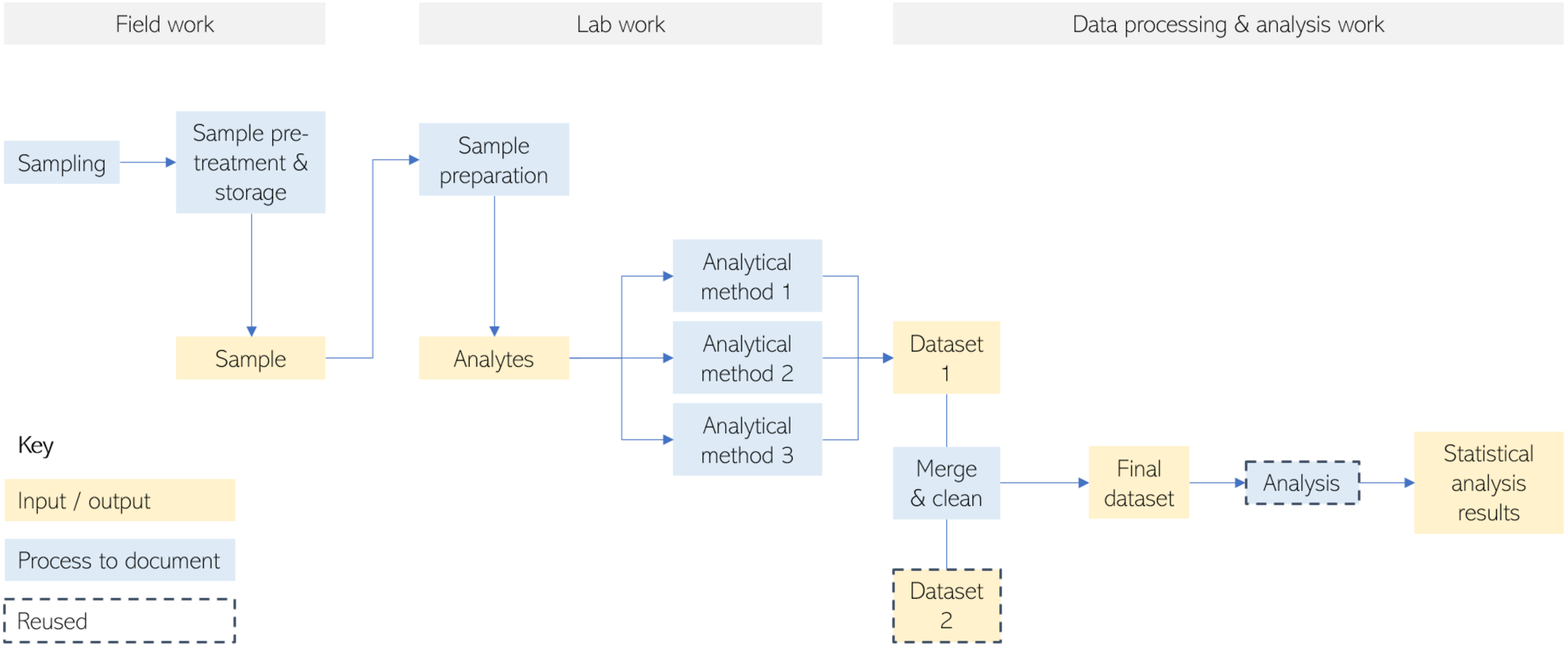
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DATA DESCRIPTION – SOME TIPS

- Only provide answers relating to research data, not publications!
- Remember that 'research data' come in many different forms
- We will start by creating a project data workflow
- Clear distinction between re-used and new data.
- Break down and list your data types conveniently: e.g. by technique, by purpose, by research phase.
- Provide enough details for outsiders to understand the sort of data involved
  - e.g. distinguish between digital/non-digital data, quantitative/qualitative data, raw & processed data
  - specify file formats (preferably no proprietary/unusual formats)
  - specify data collection methods (e.g. experimental, observational, simulation… data)

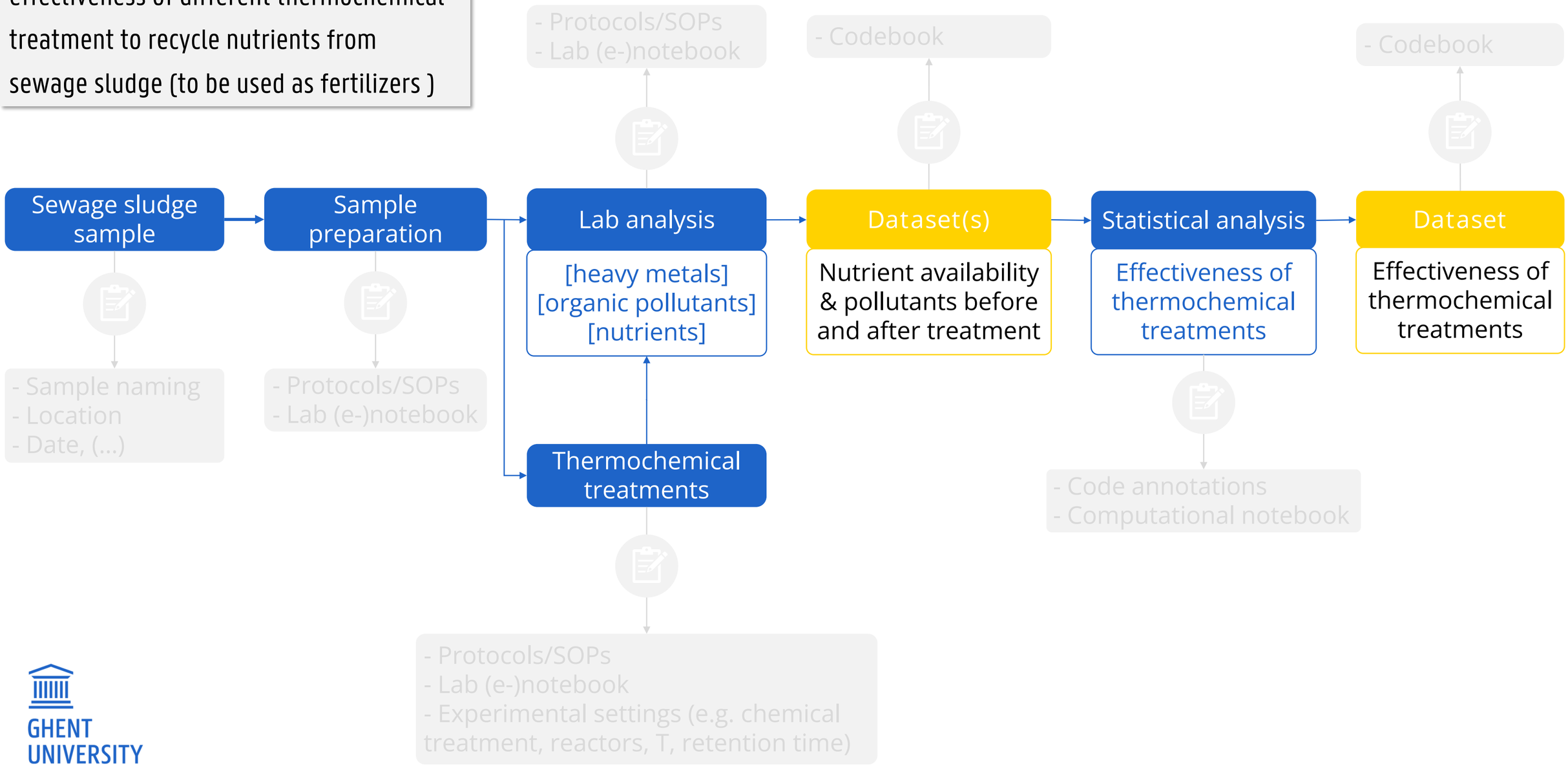
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DATA WORKFLOW: GENERIC EXAMPLE



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Research objective: investigate effectiveness of different thermochemical treatment to recycle nutrients from sewage sludge (to be used as fertilizers)



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HANDS-ON: DESIGN YOUR OWN DATA FLOW

- Paper and pen
- Look at your current DMP/project and make a flowchart representing how data flows throughout the project
  - > any missing steps?
- Identifying datasets to be managed/ undocumented or derived datasets that need to be included in the DMP.



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FROM DATA FLOW TO DATA SUMMARY TABLE

Dataset name	Description	New or reused	Data type	Data format	Volume
WP1 Pollutant concentration & nutrient bioavailability before and after treatment	Comparison of the concentration of heavy metals, organic pollutants and nutrient bioavailability, per sample, before and after different thermochemical treatments	New	Experimental, quantitative	xlsx, csv	<5MB
WP1 Data from statistical analysis	Code and results from statistical analysis to determine the influence of the dependent variables (chemical treatment/reactors, temperature, retention time) in pollutant removal and nutrient bio-availability.	New	Analysis script & results, quantitative	R, csv, png	<20MB
WP2 Consumers attitudes - questionnaire responses	Data on attitudes towards consumption of food produced with fertilisers based on recycled human excreta. Online questionnaires will be performed using Qualtrics and will collect: ● demographic data from participants: place of residence, age, gender, education, occupation, annual income. ● respondent's opinions or views about issues in relation to the research topic: environmental awareness questions, attitudes or acceptance of the proposed product, etc.	New	Observational, quantitative & qualitative	csv	<5MB
WP2 Consumer attitudes - analysis	Statistical analysis of survey responses	New	Analysis results, quantitative	sav	<10MB

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DATA SUMMARY TABLE: TIPS

- How much granularity is needed?
  - E.g. How do I break down the data into different rows?
  - Project dependent. Which characteristics are useful or create specific RDM needs?
    - Collection method, content type, WP…

- How much details should I provide?
  - E.g. How many columns or characteristics should I provide?
  - DMP templates-> table template or explicit requirements, e.g. method, format and expected size.
  - Add other relevant characteristics that are specific to your data and that have implications for data management.
    - E.g. sensitive or not, temporal/geographical scope…

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HANDS-ON: CREATE/COMPLETE YOUR DATA SUMMARY TABLE

- Fill in the data summary table provided
  - Depart from your data workflow chart
    - Update your DMP with actual data, formats, tools, etc.
  - Clear distinction between re-used and new data.
  - Break down and list your data types conveniently: e.g. by technique, by purpose, by research phase.
  - Provide enough details for outsiders to understand the sort of data involved



Dataset name/ reference	**Brief description**	**Data type** audiovisual - numerical - image - model - software - text - workflow - physical object - other	**New or reused**	**Format**	**Volume** <100MB <1GB <100GB <1TB <5TB <10TB <50TB >50TB NA

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RDM CHECK

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WHAT ARE POSSIBLE RED FLAGS FOR DATA MANAGEMENT?

- What will have an impact on the way you can manage data?
- … on what you can do with data?
- What can cause obstacles during or towards the end of the project?
- …

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CHECKLIST

- Ethical approval needed
- Privacy, processing personal data
- Clinical studies: regulations
- Third party collaborators
- Reuse of data: contracts? Licenses?
- Valorization
- Large data volumes
- …

TASK: Add stickers in the data summary table for those datasets that will require special attention when planning data management

Dataset name/ reference	**Brief description**	**Data type** audiovisual - numerical - image - model - software - text - workflow - physical object - other	**New or reused**	**Format**	**Volume** <100MB <1GB <100GB <1TB <5TB <10TB <50TB >50TB NA

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DATA SUMMARY TABLE: RDM CHECK

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WP2 Consumer attitudes - analysis	Statistical analysis of survey responses	New	Analysis results, quantitative	sav	<10MB

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SECURELY STORING DATA DURING PROJECTS

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DATA STORAGE & SECURITY QUESTIONS

- Where is the data stored?
- How is the data backed up?
- How do you ensure that the data are securely stored and not accessed or modified by unauthorized persons?
- When working with personal or otherwise confidential data, explain the pseudonymization procedures and other security measures (e.g. encryption, multi-factor authentication, etc.).

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THE DO'S AND DON'TS OF DATA SECURITY

Avoid local storage or own IT resources

Avoid storing confidential information in third-party cloud services

Do not send data by email

Use institutional storage

Backup regularly

Configure access permissions

Encrypt confidential data

Use safe file transfer alternatives

Keep your system updated

Use strong passwords

Protect your hardware

Control physical access to confidential data

When needed, destroy information in a safe way

- Shred paper
- Erasing software
- Destroy hardware

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More: https://www.ugent.be/en/research/datamanagement/during-research/data-security.htm

Knowledge clip: Data security

RECOMMENDED STORAGE OPTIONS OFFERED BY UGENT



- **SharePoint sites:** cloud-based storage offered by UGent DICT
- Large **capacity** but individual file size <250MB
- Easy to **synchronize** to local: accessible anywhere via browser or from file explorer (also offline)
- Roles & permissions to control access.
  - Give access to (at least) your promotor
    - (vs. OneDrive) Avoid data is lost when/if your account is suspended
- Automatic **back-up and version control**
- **RDM workspace template** (How to request)
- More info & demo recording



- **Shares:** network storage offered by UGent DICT. Server in S10
- Also known as *vakgroep* shares, project shares, ACL shares, UGent central disk space…
- Large capacity, no file size limit
- Need to be connected to UGent network: work on campus or with VPN
- No offline work
- Automatically backed-up: easy to recover files
- How to request
- More info & demo recording

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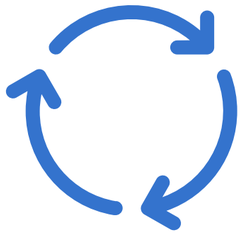
WHAT IF…?

- **Local storage is inevitable** (pc, laptop, USB, external hard drive…): field work, device or instrumentation limitations…
  - Higher risk of data loss
    - Make back-ups and make them happen automatically: sync software (e.g. FreeFileSync)
  - Higher risk for confidential data
    - Account password does not protect data/files on laptop's memory disk
    - Encrypt!
- You work with very large data -> is HPC an option?
- You work with your department solution: get informed! (Quota? Back-ups? Guidelines?)
- You need more than one storage platform:
  - Avoid when possible
  - Make clear arrangements: what goes where

CONFIDENTIAL DATA

Data for which the unwanted disclosure will have negative or harmful consequences

(may be unlawful, put participants at risk, put business processes at risk)…

RISKS  MEASURES

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POSSIBLE WORKFLOWS

Sensitiveness of data & security measures

Main storage location	Back-up	Access control	Encryption
SharePoint site	Auto Sync to local As extra layer	Configurable (roles & permission)	Encrypt local copies
Shares (network drives)	Auto As extra security layer	Configurable (ACL lists)	As extra security layer
Local storage	Manual Sync to safe location (shares) As extra security layer	Not supported	Encrypt local copies

WHERE IS YOUR DATA STORED?

- Fill in the storage column of the DMP
  - Where will you store your data?
  - What is the back-up strategy?
  - Who will have access to the data?
  - What other security measures will you implement (e.g. encryption, passwords and MFA)?



Dataset name/ reference		Where is data stored? How is it backed-up?	Who can access? How is access control implemented? Other security measures?

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BREAK

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DATA SHARING

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DATA SHARING & REUSE QUESTIONS

- Specify for each dataset how it will be made findable or discoverable by others: access level, access conditions, location. If you cannot share the (meta)data, clarify the reason for this.
- When will the data be made available? For each dataset, state when it will be made findable and if an embargo is foreseen.
- If already known, please specify the data usage licenses that you are going to assign to each dataset.
- For each dataset deposited in a repository, mention the persistent identifier (e.g. DOI, accession number, …) provided by that repository.
- What are the expected costs for data sharing? How will these costs be covered?

- Do you have to share data?
- Can you share data?

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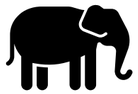
RESTRICTIONS ON DATA/SOFTWARE SHARING



Personal data



Confidential data



Sensitive data



Third-party data



Copyright

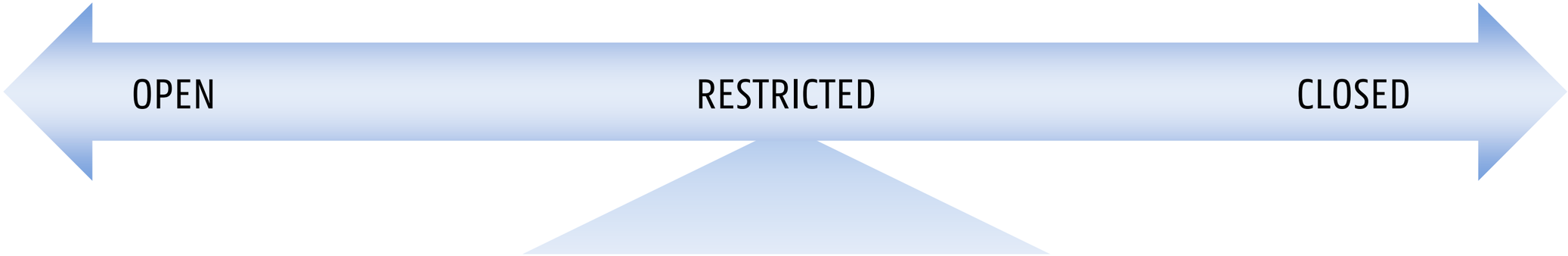


Valorization

More info at: https://www.ugent.be/en/research/datamanagement/after-research/sharing.htm#Restrictionsondatasharing

WHAT ABOUT SHARING OF SENSITIVE OR CONFIDENTIAL DATA?

"As open as possible, as closed as necessary"



Can be freely used, modified & shared by anyone for any purpose

- Personal data
- Confidential data
- Sensitive data
- 3rd party data
- Copyright/database right
- Valorization potential

Under embargo
Unable to share

QUESTION: where do (each of) your data fit?

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PRACTICAL: WAYS OF SHARING DATA



Upon request



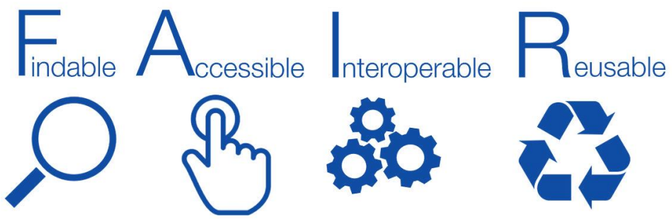
Personal or project website



Article suppl. material



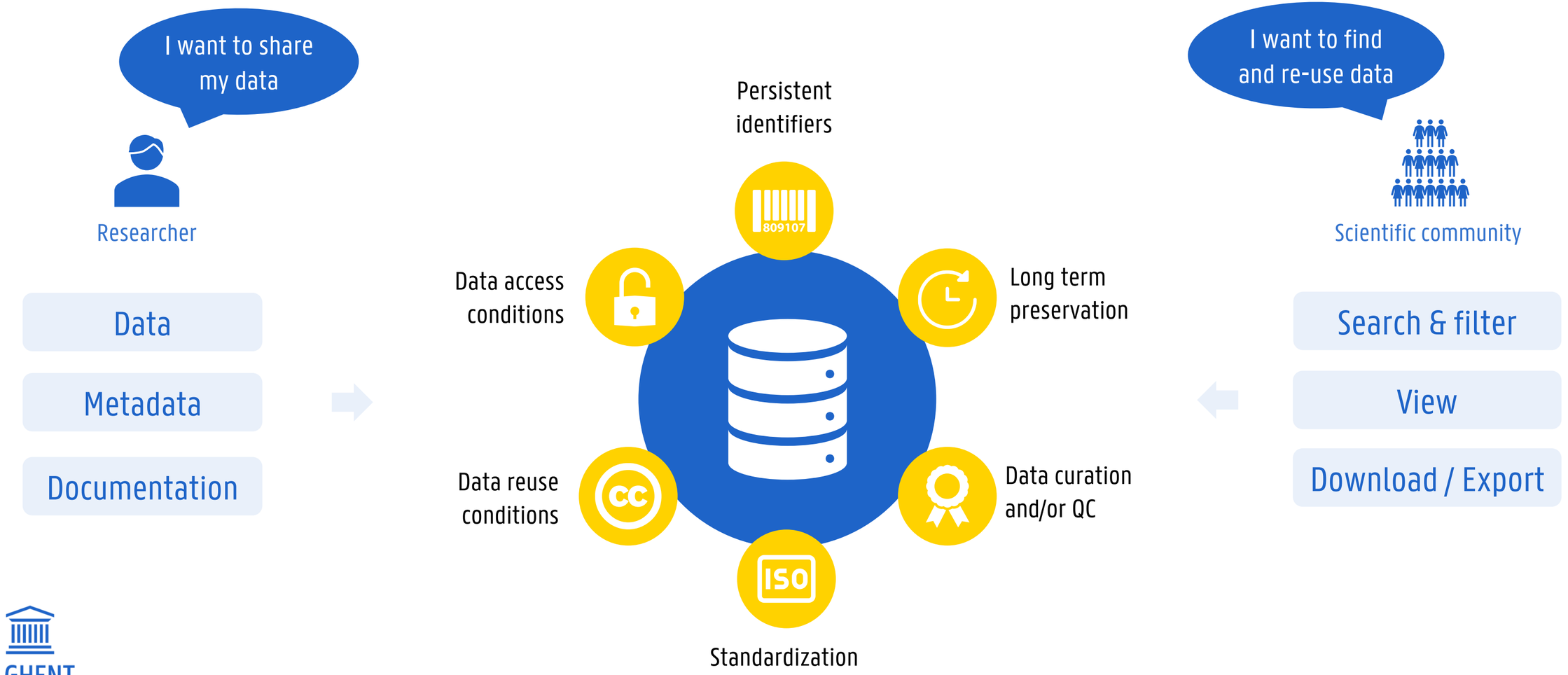
Data repository or archive



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WHAT IS A DATA REPOSITORY?

A database infrastructure that collects, manages and stores data for preservation and sharing



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DATA REPOSITORIES & FAIR DATA

→ Key infrastructures that provide functionalities to make your data FAIR.

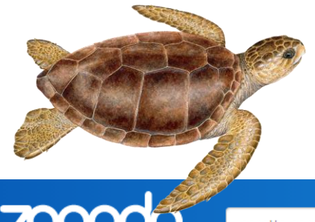
→ Shared responsibilities between repositories and researchers in making data FAIR

FAIR principle	In practice	Repository	Researcher
FINDABLE	Assign a persistent identifier (PID)	●	
	Data are described with rich metadata	●	●
	Metadata include the PID, are indexed & searchable online	●	
ACCESSIBLE	(Meta)data retrievable via PID using a standardised communications protocol (e.g. html)	●	
	Authentication and authorisation procedure	●	
	Metadata remain accessible (even if data aren't)	●	
INTEROPERABLE	Metadata standards	●	
	Controlled vocabularies	●	●
	References to other (meta)data	●	●
REUSABLE	(Machine readable) license / terms of use	●	●
	Detailed provenance documentation		●
	Domain standards	●	●

GENERIC VS. DOMAIN REPOSITORIES

- Domain-specific repositories
  - Visibility: where your peers will likely look for data
  - Use of discipline standards & vocabularies
    - Search filters, visualization tools tailored to domain
    - Interoperability: easier to integrate multiple datasets
  - Dataset curation by repository advisors often part of submission process
- Generic repositories:
  - Provide rich metadata to make it more findable & reusable
- Generic AND domain:
  - DOI, link to publication(s), dataset(s) or other outputs
  - Support minimum degree of FAIRness

GENERIC VS. DOMA



zenodo | caretta caretta | Upload | Communities

All versions

Found 41 results.

Access Right
- Open (27)
- Closed (14)

File Type
- Pdf (18)
- Png (4)
- Xlsx (3)
- 7z (1)
- Html (1)
- Xml (1)
- Zip (1)

Keywords
- Biodiversity (22)
- Taxonomy (22)
- Animalia (15)

November 12, 2019 (v1) | Journal article | Closed Access

The widely occurring brittlestar Ophiactis savi… loggerhead sea turtle, Caretta caretta

Hayashi, Ryota; Okanishi, Masanori;

Hayashi, Ryota, Okanishi, Masanori (2019): The widely occur… on loggerhead sea turtle, Caretta caretta. Zootaxa 4695 (5)…

Uploaded on November 15, 2019

Published in Zootaxa, vol. 4695, issue 5, pp. 497-500.

December 31, 1758 (v1) | Taxonomic treatment | Open Access

Testudo caretta Linnaeus, 1758, spec. nov.

Linnaeus, Carolus;

Testudo caretta [spec. nov.] T. pedibus pinniformibus, unguibus palmarum plantarumque binis, testa ovata acute serrata. … 85. n. 69. Testudo pedibus natatoriis, unguibus acuminatis binis. Brown. jam. 465. Testudo unguibus utrinque binis acutis, … quinque gibbis. C

Uploaded on June 29, 2020

October 24, 2020 (01) | Dataset | Open Access

Relativistic Kinematics of a Magnetic Soliton: Manuscript Data

Caretta, Lucas; Oh, Se-Hyeok; Fakhrul, Takian; Lee, Dong-Kyu; Lee, Byung Hun; Kim, Se Kwon; Ross, Caroline; Lee, Kyung-Ji… Geoffrey;

"*Caretta caretta*" in Zenodo

Occurrences

Search all fields

Simple | Advanced

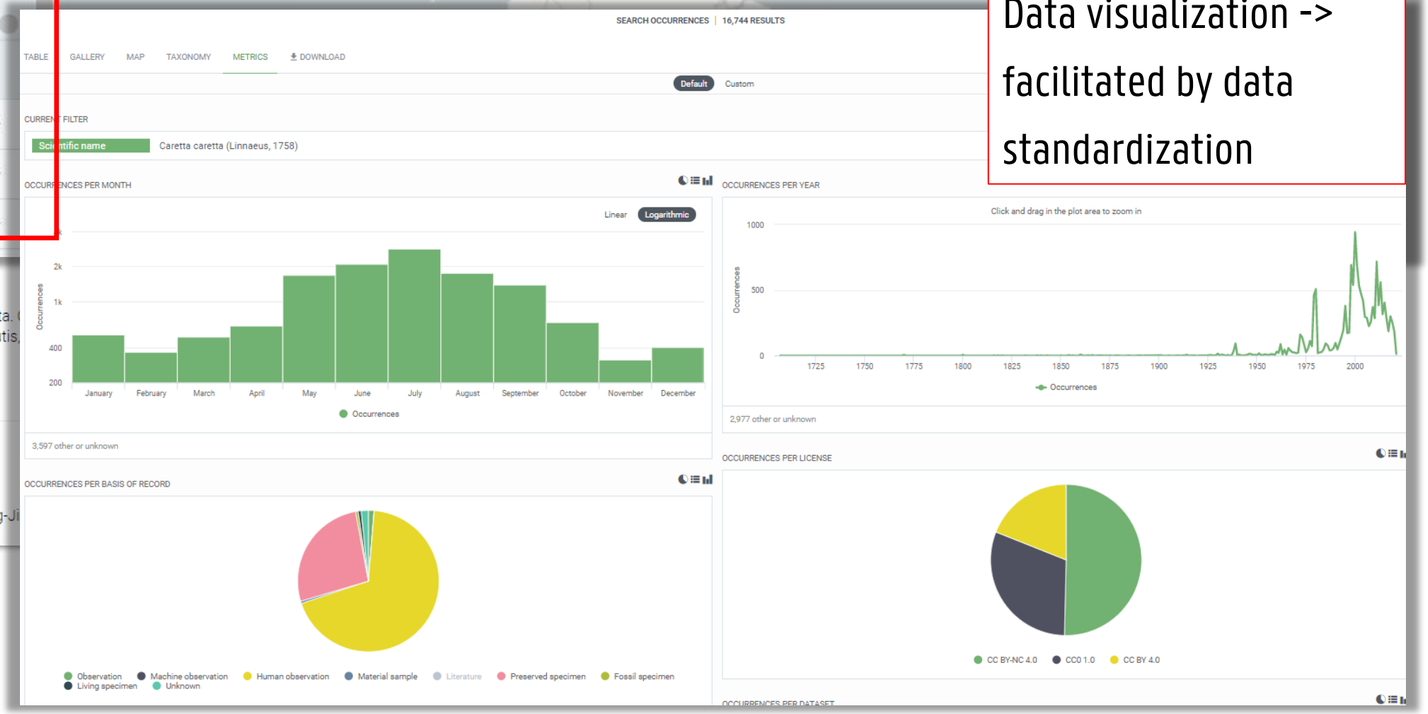
- Occurrence status
- License
- Scientific name
  - Caretta caretta (Linnaeus, 1758)
- Basis of record
- Location
- Administrative areas (gadm.org)
- Coordinate uncertainty in meters
- Year
  - No filter
  - 1000 – 2021
- Month
- Dataset
- Country or area

Search filters tailored to domain & data type -> facilitated by domain metadata standard

SEARCH OCCURRENCES | 14,390 WITH COORDINATES

TABLE | GALLERY | MAP | TAXONOMY | METRICS | DOWNLOAD

"*Caretta caretta*" in GBIF



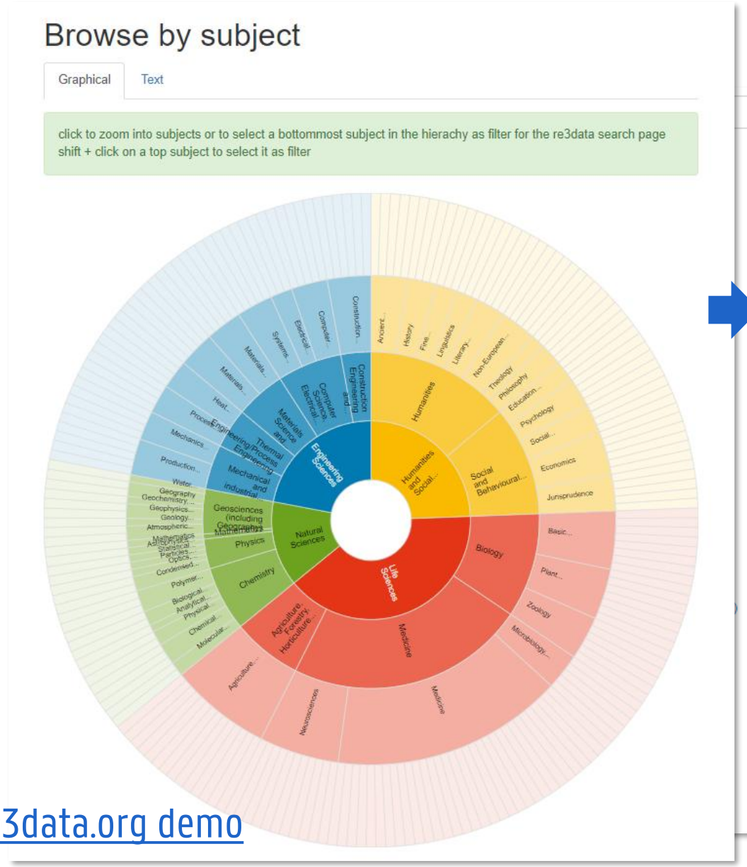
Data visualization -> facilitated by data standardization

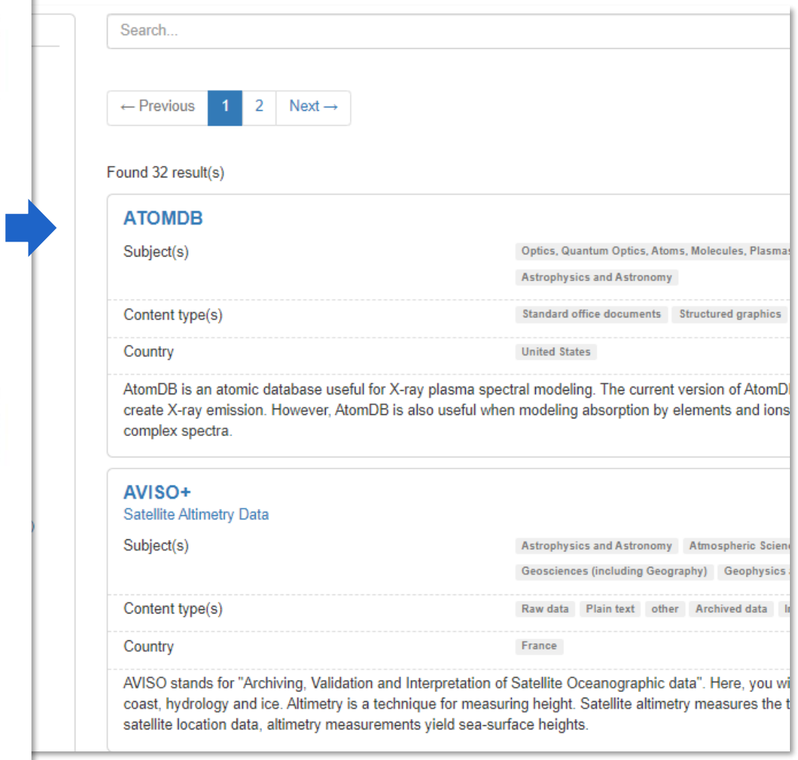
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Global Biodiversity Information Facility

HOW TO SELECT A DATA REPOSITORY?

1. Where do your peers publish data? Domain's consensus
2. Does the journal / funder recommend or require certain repositories?
3. Registries of data repositories: re3data.org, FAIRsharing





re3data.org demo



Data repositories recommended by Nature

LICENSES

- Licenses: means for owners of rights (in datasets) to **grant permission(s).**
- Define what can and cannot be done with data in a clear way.
  - Nature: expand rather than restrict what can be done.

    E.g, data may be 'publicly available' but, without license, users do not have permission to access, use and share it under copyright or database laws.
- Licenses are **not automatic.**
- Avoid custom-made licenses / agreements as much as possible (not always possible).
- Standard licenses: terms and conditions are already defined.
  - Creative Commons, Open Data Commons, open software licenses

HANDS-ON: DATA SHARING & REUSE

- Have you identified potential repositories?
- What access level would be assigned to your dataset(s)?
- If restricted data, what access conditions? Any suitable repository?



Dataset name/ reference		Where are you sharing data? Access level? For open data, which license. Not open data, indicate why + access level, access conditions

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PRESERVATION

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PRESERVATION QUESTIONS

- Indicate for each dataset how long it will be preserved after the research project is finished, and at which storage location. If you cannot preserve the data for at least five years (or longer, in agreement with other preservation policies that are applicable), clarify the reason for this.
- What are the expected costs for data preservation? How will these costs be covered?

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PRESERVING VS STORING

- Preserving data
  - completed research
  - data (or milestone versions of data) that are ‘frozen’ and not in active use.

- Actions to prevent data from becoming unavailable and unusable over time, for example because of:
  - ‘Owners’ leaving
  - Outdated software or hardware
  - Storage media degradation
  - A lack of sufficient descriptive and contextual information to keep data understandable

Data preservation

≠

Not deleting the data files

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WHERE TO PRESERVE DATA?

- Research data and documentation selected for retention should be kept in a **suitable location**
  - in a secure manner to ensure that they remain available and usable beyond the end of your project
  - with appropriate access rights,
  - and in sustainable formats.
- Suitable location? -> main purpose of preservation (verification, internal or external reuse).

Data repository

- Recommended / preferred practice.
- *The* option **when goal is sharing & reuse**.

In house

- When not possible/appropriate to deposit data externally, e.g. for legal, ethical, contractual, practical, or other reasons.
- Data needs to be preserved for verification purposes
- In-house options: Project Shares.

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HANDS-ON: PRESERVATION

- Where will data be preserved?
- How long will data be preserved?



Dataset name/ reference		Preservation location & period

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DOCUMENTATION AND METADATA

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DOCUMENTATION & METADATA QUESTIONS

- What documentation or other supporting material will accompany the data for it to be interpreted correctly? e.g. (Electronic) Lab Notebooks, README files, codebooks etc
- How will metadata be provided for the data? If already known, list for each dataset which standard metadata schema will be used and/or what type of metadata will be provided.
  - Provided in the data file
  - Via a data storage or management platform
  - Via a data repository
  - As a separate file accompanying the data

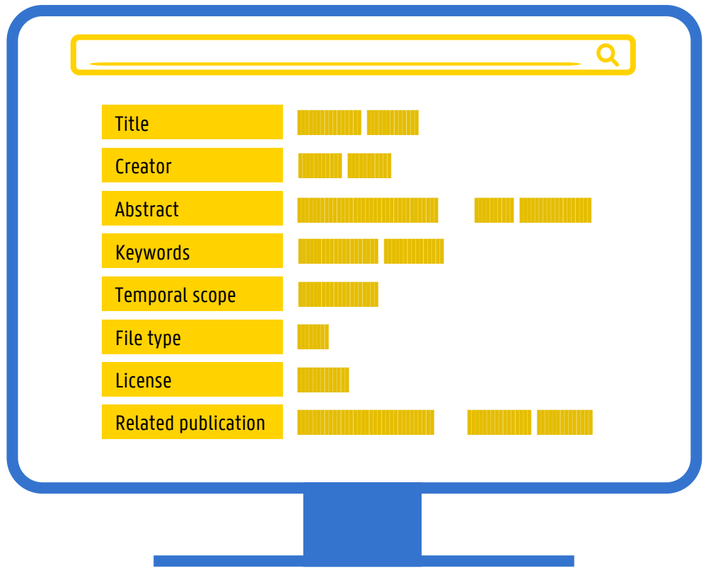
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DOCUMENTATION VS. METADATA

DOCUMENTATION: any contextual information that allows data to be understandable and reusable.

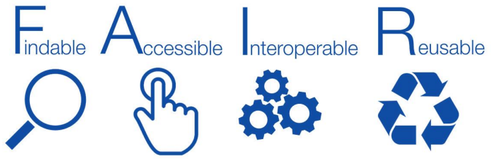


METADATA: information about data, recorded in a **highly structured** way, using a set of defined information fields or elements to be **machine readable**.





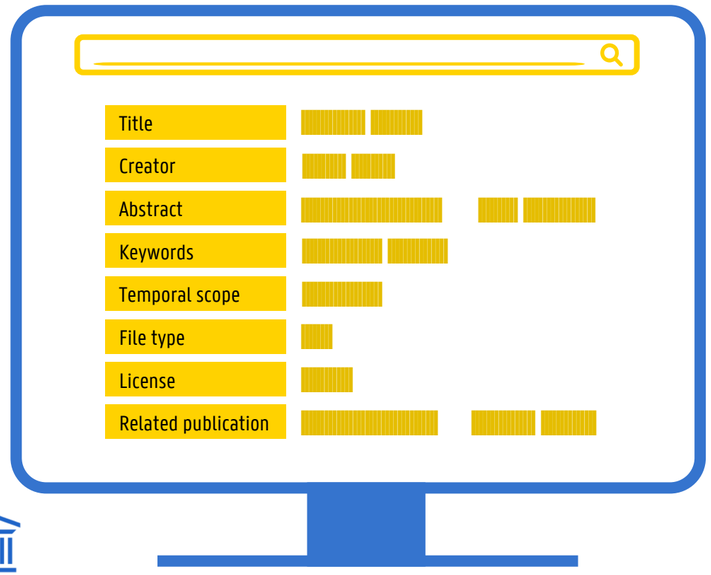
- Metadata allows others:
  - To find data
  - To assess whether it is useful
  - To know whether they can reuse it, and under which conditions (license), and how it can be accessed
- Documentation allows others:
  - To understand where the data came from and how to (re)use it

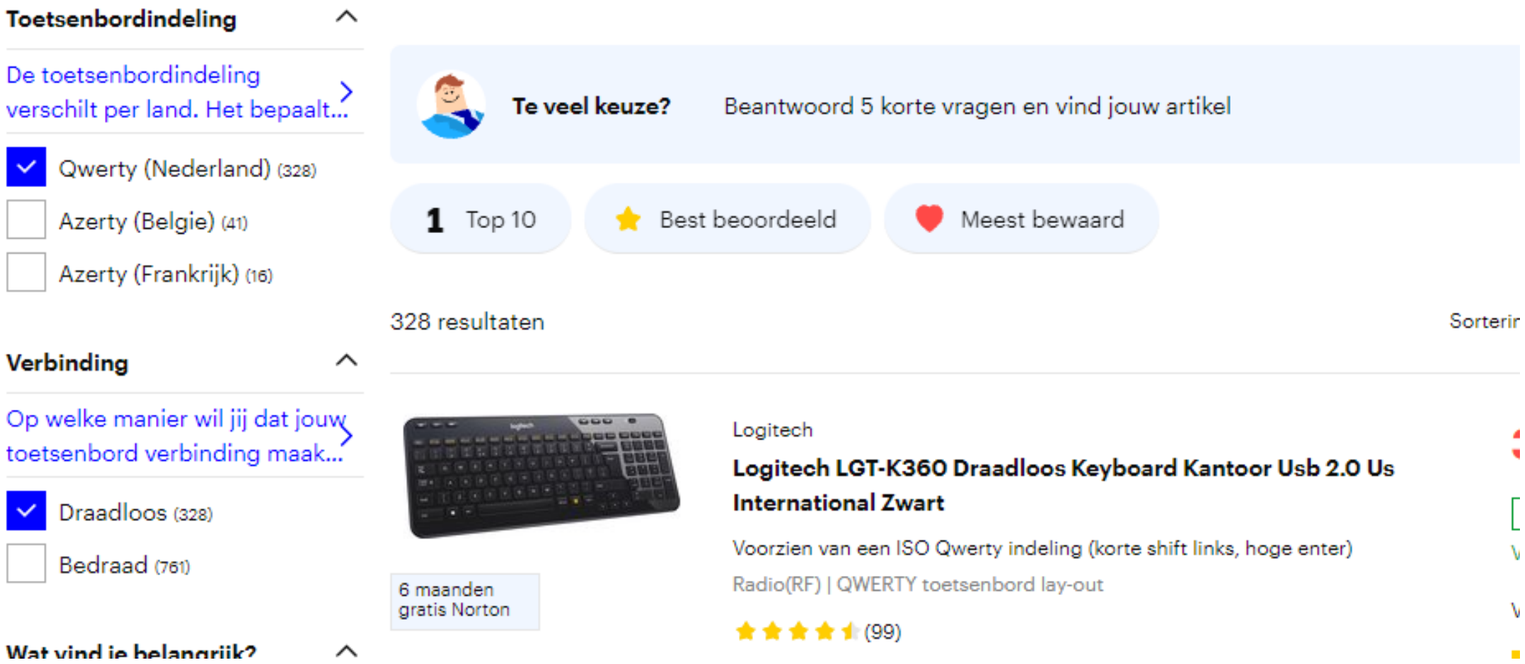


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METADATA BASICS

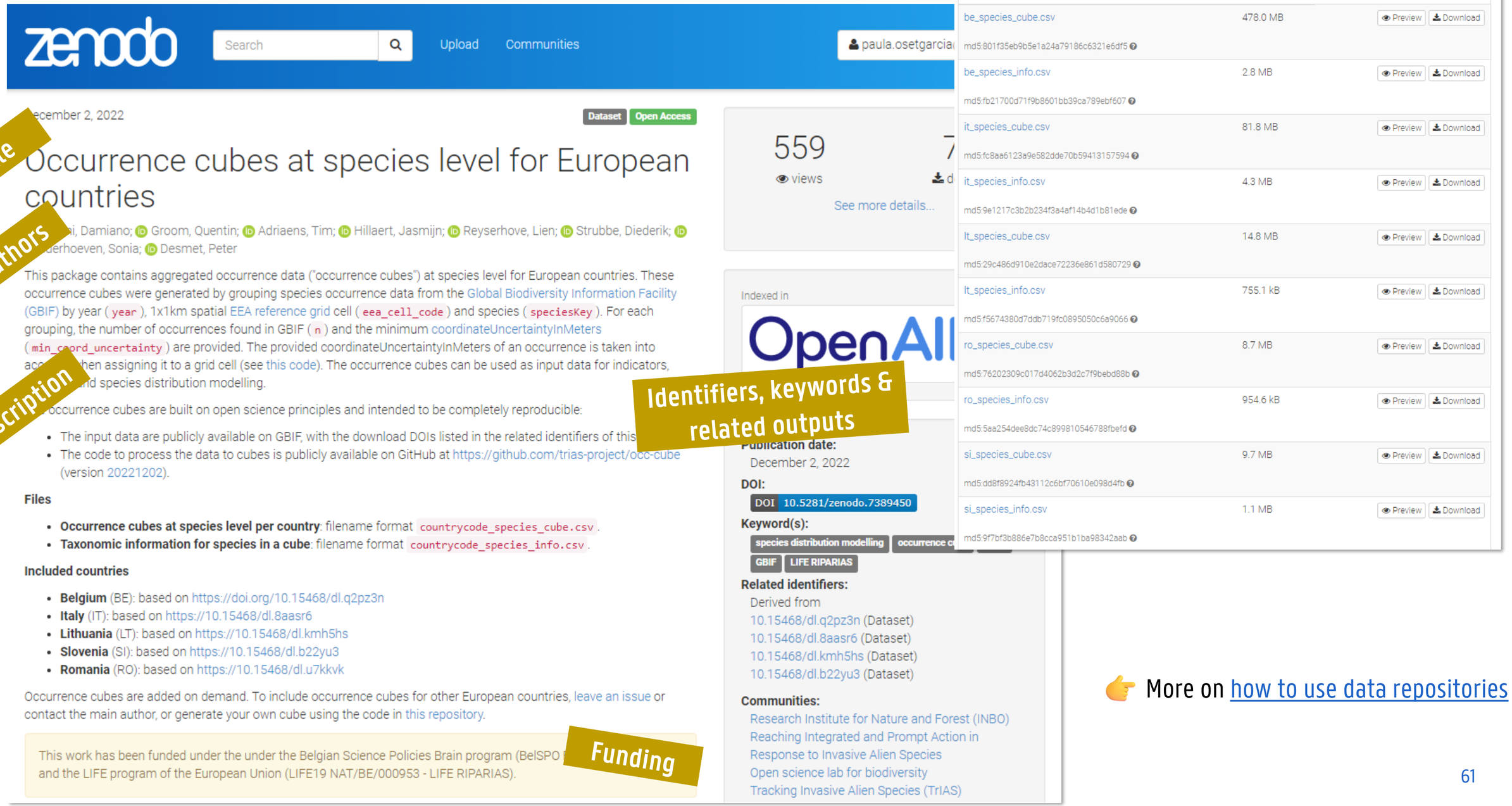
- Information about data (or any digital object)
- A (very) structured type of documentation: records information in sets of **pre-defined fields or elements**.
- Meant to be read & **exchanged by machines**
- To **facilitate (online) data discovery**
- To quickly judge usefulness of data





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DATA REPOSITORY (ZENODO): METADATA



Title

Authors

Description

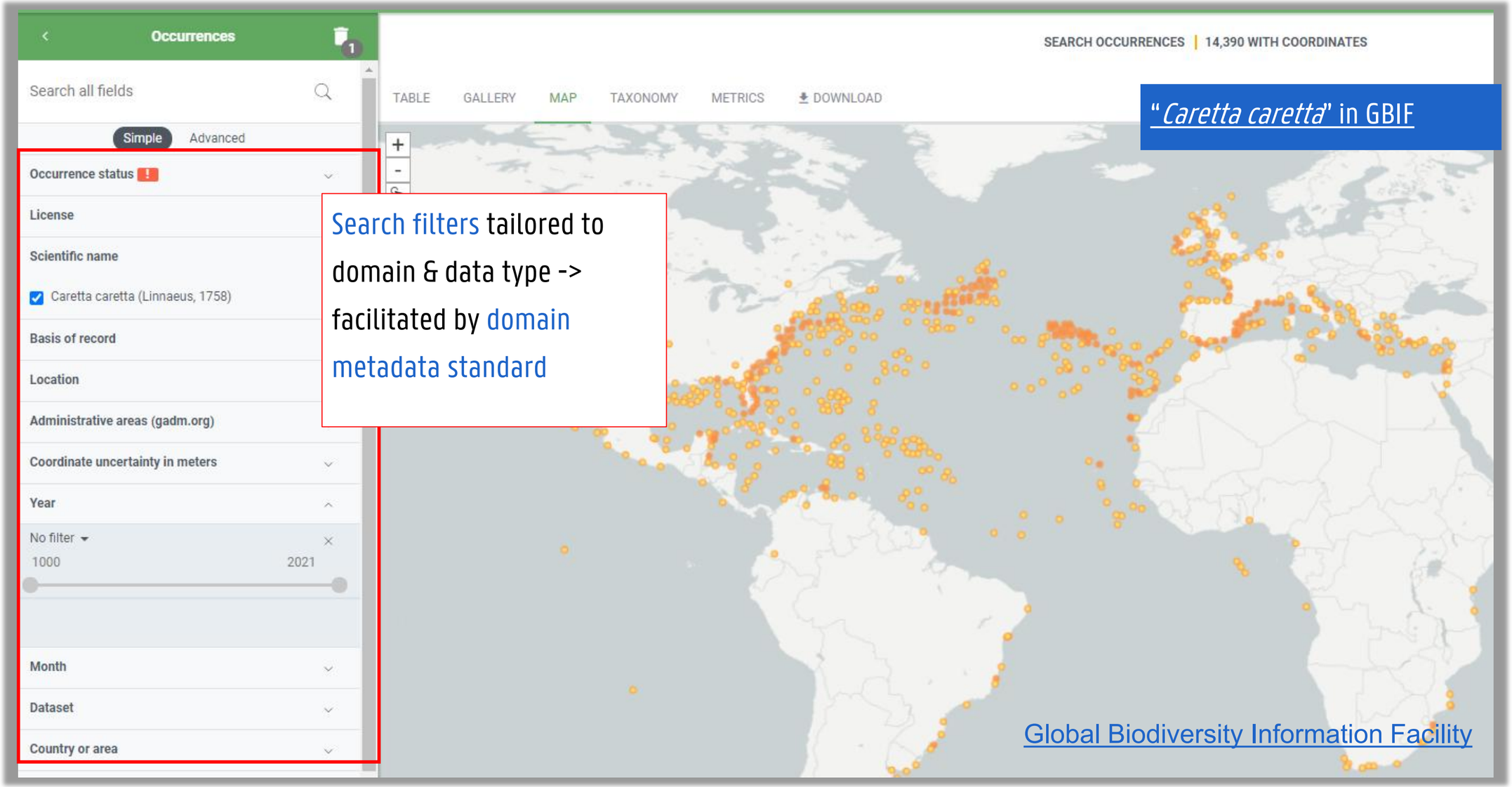
Identifiers, keywords & related outputs

File names & sizes

Funding

👉 More on how to use data repositories

EXAMPLE OF METADATA IN A DOMAIN-SPECIFIC REPOSITORY



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WHICH METADATA (STANDARD)? CHECK REPOSITORY GUIDELINES

Creating metadata for a data package

Metadata are data that describe the structure and context of other data. They are vital to the discovery and reuse of a dataset, and are a required element of an EDI data package. The EDI Data Repository operates on the EML metadata standard, a widely adopted and actively maintained metadata specification. For more guidance on how to create EML metadata and associated best practices see Creating Metadata for Publication.

https://edirepository.org/support/frequently-asked-questions
https://edirepository.org/resources/creating-metadata-for-publication

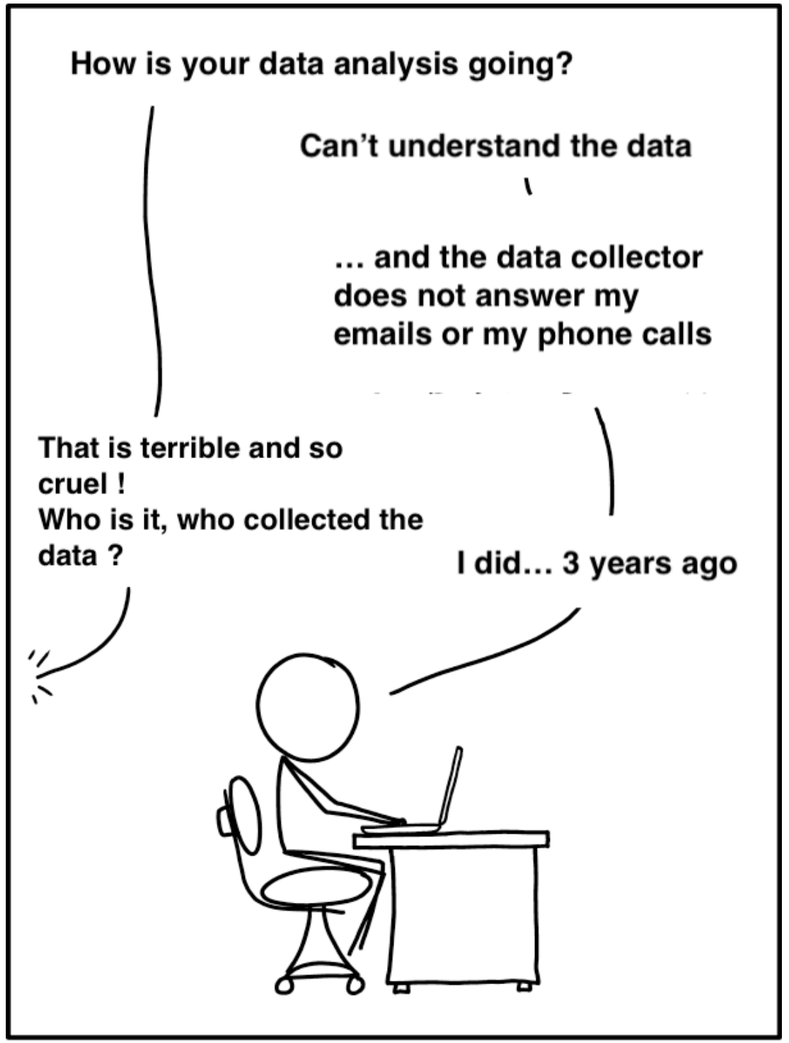
To be Findable:

- **F1**: (meta)data are assigned a globally unique and persistent identifier
  - A DOI is issued to every published record on Zenodo.
- **F2**: data are described with rich metadata (defined by R1 below)
  - Zenodo's metadata is compliant with DataCite's Metadata Schema minimum and recommended terms, with a few additional enrichments.
- **F3**: metadata clearly and explicitly include the identifier of the data it describes

https://about.zenodo.org/principles/

WHY DOCUMENTING DATA?

- Help yourself & others:
  - “Your future self will be your first collaborator”
  - Will you remember or will others understand…
    - what your variables mean?
      - E.g. "sglmemgp" = single member of group
    - the procedures used to transform or derive variables?
  - Keep data understandable & reusable
- Help science:
  - Research review & verification
  - The reproducibility crisis



Your first collaborators are your future selves, be nice to them !

your future self, by Julien Colomb, CC-BY-NC, derived from .NORM Normal File Format, CC-BY-NC, by Randall Munroe

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CODEBOOK EXAMPLE

- Covid data: https://epistat.wiv-isp.be/covid/
- https://epistat.sciensano.be/COVID19BE_codebook.pdf

COVID19BE_VACC_MUNI_CUM

CUMULATIVE NUMBER OF ADMINISTERED VACCINES BY WEEK, MUNICIPALITY, AGE, AND DOSE

Variable	Definition
YEAR_WEEK	Week (Monday to Sunday) including all vaccines that were administered by the end of that week and registered in Vaccinnet+
NIS5	NIS-code (municipality) of residence of people receiving the vaccine (5 digits)
AGEGROUP	Age group of people receiving the vaccine
DOSE	Dose identifier (for vaccines requiring 2 doses: A for first dose, B for second dose; C for vaccine requiring only 1 dose; E for extra/booster dose of vaccine administered since week 36 of 2021)
CUMUL	Cumulative number of people receiving the vaccine (combinations with less than 10 vaccinations are indicated with “<10”, combinations with 0 vaccination are not shown)

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README FILES

This DATSETNAMEreadme.txt file was generated on YYYY-MM-DD by NAME
<help text is included in angle brackets, and can be deleted before saving>

GENERAL INFORMATION

1. Title of Dataset:

2. Author Information
   A. Principal Investigator Contact Information
      Name:
      Institution:
      Address:
      Email:

   B. Associate or Co-investigator Contact Information
      Name:
      Institution:
      Address:
      Email:

   C. Alternate Contact Information
      Name:
      Institution:
      Address:
      Email:

3. Date of data collection (single date, range, approximate date) <suggested format YYYY-MM-DD>:

4. Geographic location of data collection <latitude, longiute, or city/region, State, Country, as appropriate>:

5. Information about funding sources that supported the collection of the data:

DATA & FILE OVERVIEW

1. File List:
<list all files (or folders, as appropriate for dataset organization) contained in the dataset, with a brief description>

2. Relationship between files, if important:

3. Additional related data collected that was not included in the current data package:

4. Are there multiple versions of the dataset? yes/no
   A. If yes, name of file(s) that was updated:
      i. Why was the file updated?
      ii. When was the file updated?

METHODOLOGICAL INFORMATION

1. Description of methods used for collection/generation of data:
<Include links or references to publications or other documentation containing experimental design or protocols used in data collection>

2. Methods for processing the data:
<describe how the submitted data were generated from the raw or collected data>

3. Instrument- or software-specific information needed to interpret the data:
<include full name and version of software, and any necessary packages or libraries needed to run scripts>

4. Standards and calibration information, if appropriate:

5. Environmental/experimental conditions:

6. Describe any quality-assurance procedures performed on the data:

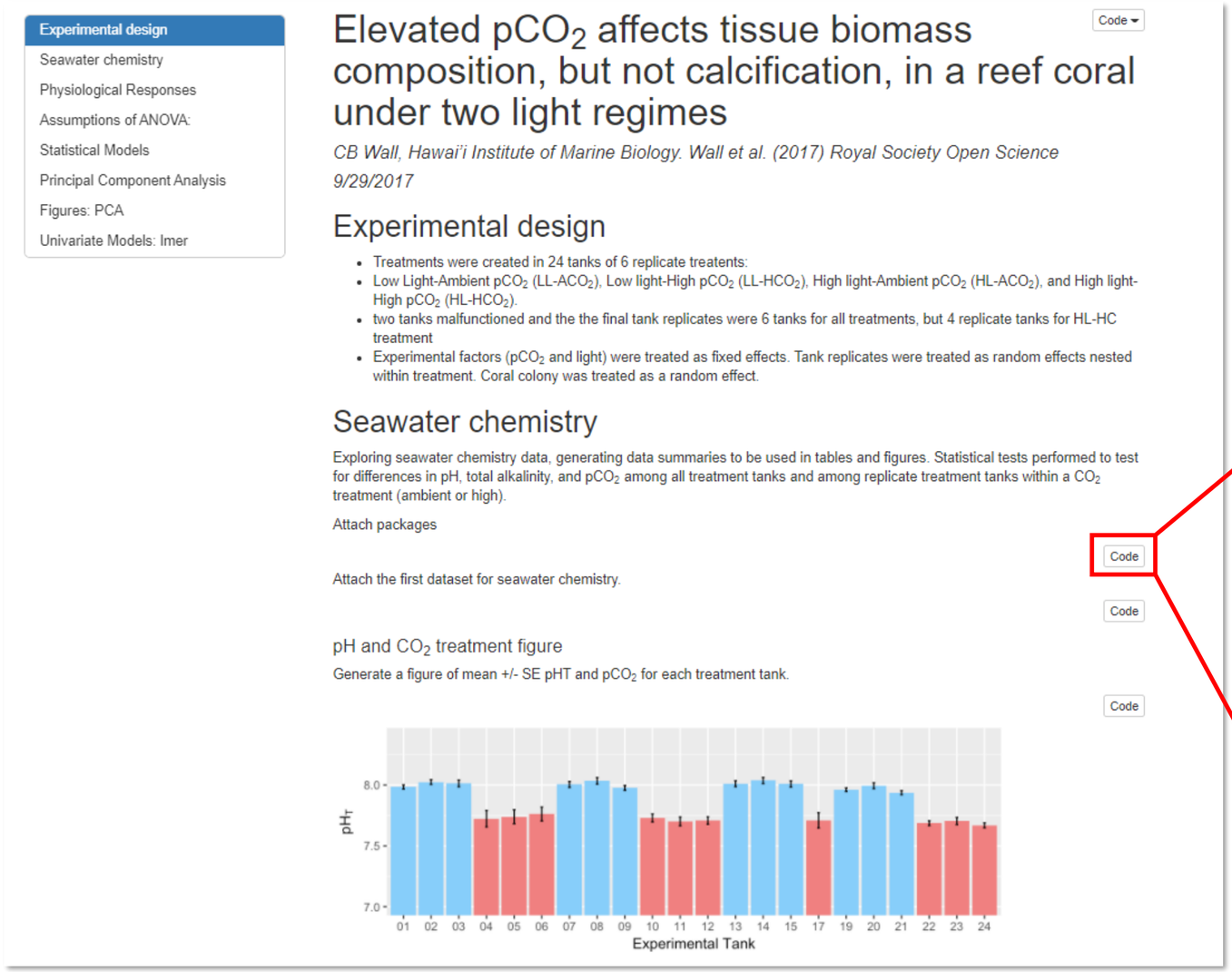
7. People involved with sample collection, processing, analysis and/or submission:

Readme template

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PROCESSING & ANALYSIS SCRIPTS

From DMP Example 2 (supplementary material):
https://doi.org/10.5061/dryad.5vg70



WHERE?

- Code annotations
- (Online) computational notebooks (e.g. Jupyter notebook, R markdown)
  - Create and share documents that contain live code, equations, visualizations and explanatory text
  - Open, reproducible research

Hide

```
# set working directory to 'figures' for figure export

setwd('markdown/figures')
########################################
## EXPERIMENT:  Figures of CO2 and pHT ##
########################################
colors=c("lightskyblue", "lightskyblue", "lightskyblue", "lightcoral","lightcoral", "lightcoral","lightskyblue", "lightskyblue", "lightskyblue", "lightcoral","lightcoral", "lightcoral","lightskyblue", "lightskyblue", "lightskyblue", "lightcoral","lightskyblue", "lightskyblue", "lightskyblue", "lightcoral","lightcoral", "lightcoral")

# plot for all tank pH and pCO2

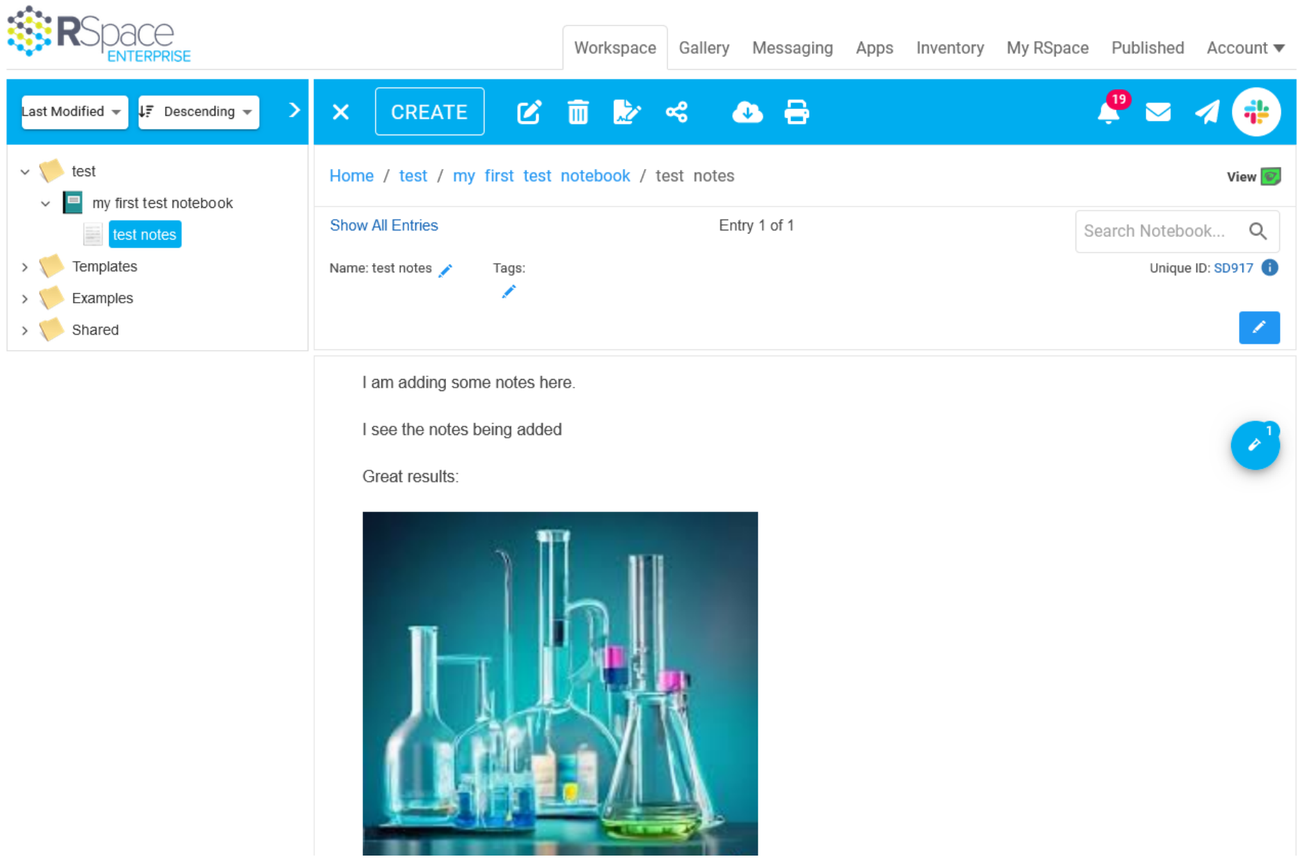
################
###### pH ######
################
mean<-tapply(SWdata$pHT, SWdata$Tank, mean); se<-tapply(SWdata$pHT, SWdata$Tank, std.error); pH<-data.frame(mean, se); pH["Tank"] <- c("01", "02", "03", "04", "05","06","07","08","09","10","11","12","13","14","15","17","19","20","21","22","23","24"); colnames(pH) <- c("mean", "se","Tank")

Fig.pH <- ggplot(pH, aes(x=Tank, y=mean)) +
  geom_bar(position="dodge", stat="identity", fill=colors) +
  geom_errorbar(aes(ymin=pH$mean - pH$se, ymax=pH$mean + pH$se), width=0.1) +
  xlab(expression("Experimental Tank")) +
  ylab(expression(pH[T])) +
  scale_y_continuous(limits=c(7.0,8.4),oob = rescale_none)

Fig.pH # gives summary figure of tank pH mean +/-SE across all tanks
```

ELN – LAB NOTEBOOK RSPACE

- Flexible notekeeping tool, incl. API availability
- Compliance GDPR, 21 CFR part 11, SOC 2 Type II…
- Suitable for multiple disciplines
- Integrated inventory management

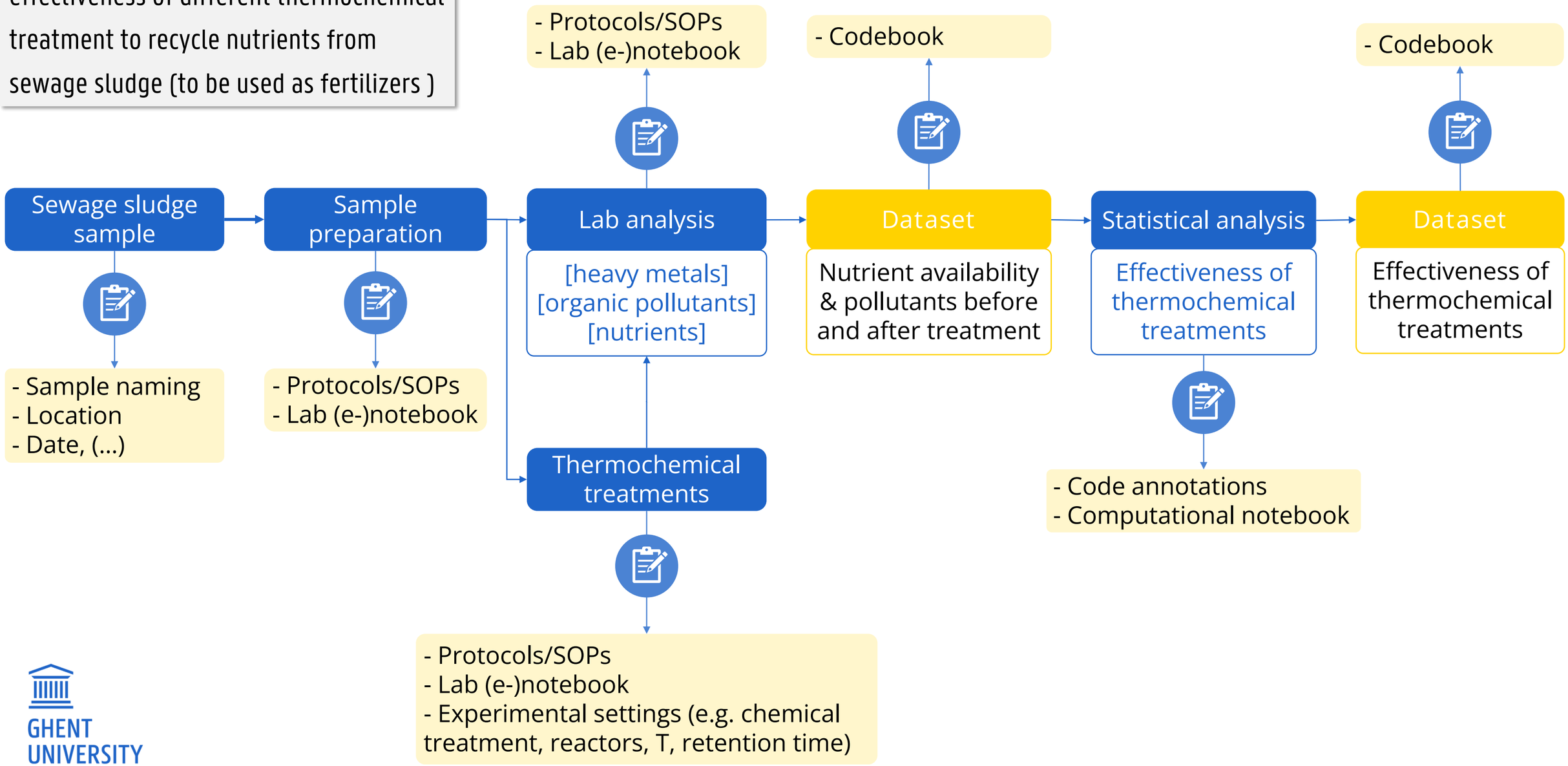


UGent setup (coming soon)

- access anywhere via browser
- login via single sign-on
- no cost for researchers
- Legal, IT security and privacy requirements covered

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Research objective: investigate effectiveness of different thermochemical treatment to recycle nutrients from sewage sludge (to be used as fertilizers)



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PROVIDING DOCUMENTATION

- Include documentation in your data package (e.g. readme file, SOPs, code notebook…)
- Include relevant links within metadata
  - E.g. link to GitHub repository
  - E.g. link to related articles and/or data paper
  - E.g. link to project website

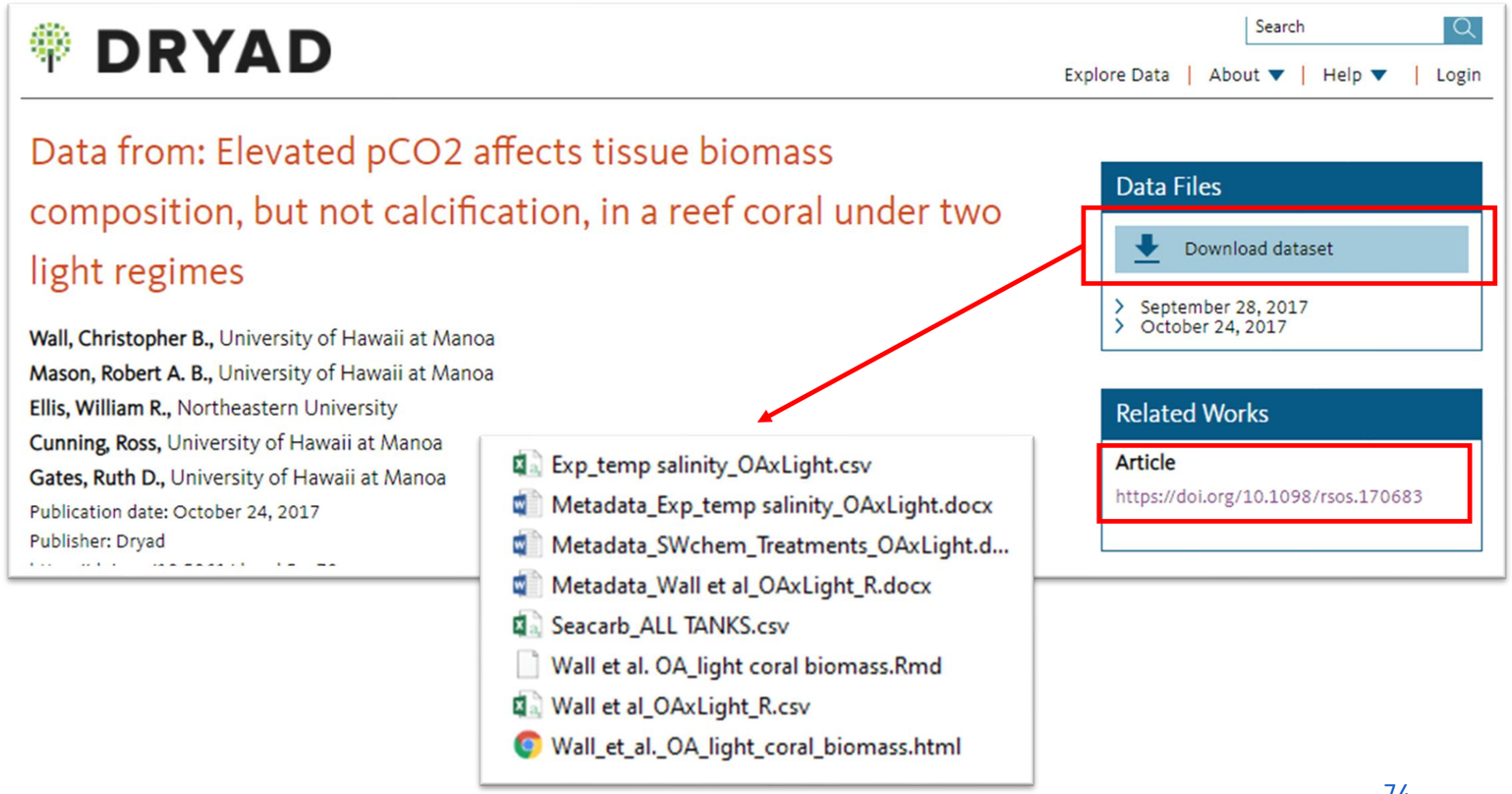
Data accessibility

All data and R code to reproduce figures and analyses are available on Dryad (doi.org/10.5061/dryad.5vg70) [51].

Footnotes

Electronic supplementary material is available online at https://dx.doi.org/10.6084/m9.figshare.c.3911941.

https://doi.org/10.1098/rsos.170683



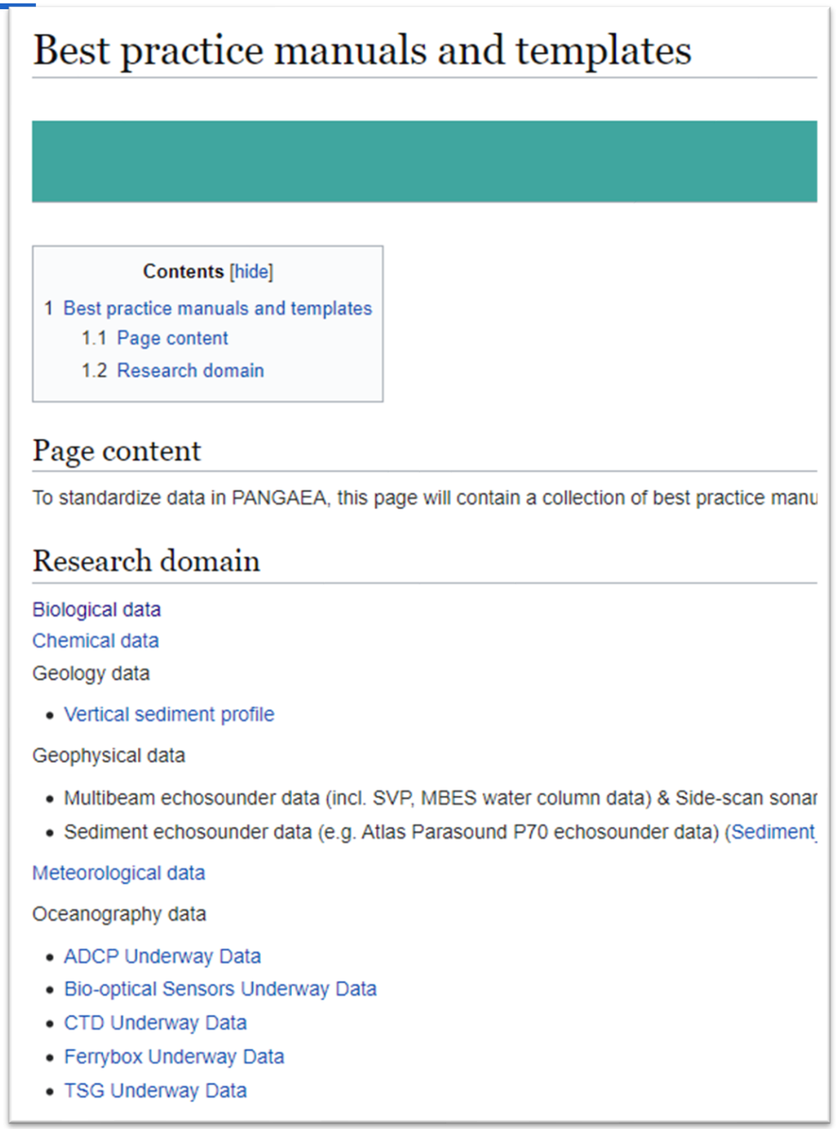
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CHECK REPOSITORY SUBMISSION GUIDELINES

Examples from PANGAEA:

- Biological data instructions
  - Template for entry of parameters, events, data
  - Dataset: https://doi.org/10.1594/PANGAEA.911931
- Geological data:
  - Template for data submission
  - Dataset: https://doi.org/10.1594/PANGAEA.922398

EDI: https://edirepository.org/resources/resources-for-information-managers



Best practice manuals and templates

Contents [hide]

1 Best practice manuals and templates
- 1.1 Page content
- 1.2 Research domain

Page content

To standardize data in PANGAEA, this page will contain a collection of best practice manu

Research domain

Biological data

Chemical data

Geology data

- Vertical sediment profile

Geophysical data

- Multibeam echosounder data (incl. SVP, MBES water column data) & Side-scan sonar
- Sediment echosounder data (e.g. Atlas Parasound P70 echosounder data) (Sediment

Meteorological data

Oceanography data

- ADCP Underway Data
- Bio-optical Sensors Underway Data
- CTD Underway Data
- Ferrybox Underway Data
- TSG Underway Data

https://wiki.pangaea.de/wiki/Best_practice_manuals_and_templates

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HANDS-ON: DOCUMENTATION AND METADATA

- Can you identify processes and/or entities to document in your data flow?
- What type of documentation can be used for each of these?
- Can you find metadata and/or documentation guidelines in the repository/repositories of your choice?



Dataset name/ reference		Documentation	Metadata (standard)

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TAKE-AWAYS

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DMP GENERAL TIPS

- Be concrete
  - Specify management actions for specific datasets/materials
- Be coherent across sections
- Be honest
  - Don’t state things that you do not intend to do
- Be pro-active
  - It’s OK not to have all answers at the beginning of the project
  - Start with the end (publishing) in mind, explore repositories and their requirements early on

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TAKE-AWAYS

- A DMP is a living document — keep it updated to match real practice.
- Data workflows help visualize and document what you actually produced. Data summary tables keet it clear for outsiders/.
- Spot red flags early and act upon.
- Good documentation = reusable data (for you and others).
- Plan for storage, sharing, and preservation before project closure.
- Support is available at all stages — contact us!

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THE BIG PICTURE

Plan for data management, keep your DMP updated

Explore potential data repositories and their requirements & guidelines.

- Inc. use of standards, vocabularies, etc.

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

FAIRsharing

DCC

Document as you go

- Notebook
- Codebook, data dictionary
- Workflows
- Protocols
- Readme files
- Data overview with metadata
- File organization and naming

Adopt domain standards during data collection & processing

Deposit data & documentation in repository; describe it with rich metadata, inc. access & reuse conditions

Register data in Biblio

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MORE INFORMATION

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ADDITIONAL RESOURCES

- https://www.ugent.be/en/research/datamanagement
- https://www.ugent.be/en/research/datamanagement/before-research/datamanagementplan.htm
- Disciplinary metadata standards: http://rd-alliance.github.io/metadata-directory/standards/
- Browse data repository by subject: https://www.re3data.org/browse/by-subject/
- DPO: dpo@uzgent.be privacy@ugent.be
- Hiruz: https://hiruz.be
- Data Science Institute (UZ only): https://intranet.uzgent.be/bad/2130/Paginas/default.aspx
- Research tips: https://researchtips.ugent.be/en/

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DOCUMENTATION RESOURCES

- READMEs
  - Guide to writing “readme” style metadata
  - README template
  - Documentation methods University of Bath
- Data Dictionaries
  - How to Make a Data Dictionary
  - Data Dictionary Template
- Codebooks
  - Codebook Cookbook
  - Sample Questionnaire with Coding
- Commented Code
  - Coding and Comment Style
- Lab/field Notebook
  - Guide for Taking Field Notes
- Metadata in practice for life sciences (by Elixir)
  - https://rdmkit.elixir-europe.org/metadata_management.html

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https://www.ugent.be/en/research/datamanagement/during-research/documentation.htm

UGent Data Stewards

RESEARCH DEPARTMENT - UNIVERSITY LIBRARY

rdm.support@ugent.be

@UGentRDM

ugent.be/en/research/datamanagement

www.ugent.be

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